

**UNIVERSITY OF ILLINOIS-CHICAGO
DEPARTMENT OF CHEMICAL ENGINEERING**

**CHE 201 – INTRODUCTION TO THERMODYNAMICS
Spring 2026**

Lecture:

M/W 9:30 - 10:45 AM EIB Room#124

Office Hours:

M/W 12.15-1.00 PM EIB Room#144 (Dr. Bilgin)

TBD (TA)

Instructor:

Dr. Betul Bilgin, bbilgin@uic.edu

Teaching Assistants:

Debjani Bhakta, dbhak@uic.edu

Discussion periods: F 10 am-10.30 am

Text:

***Fundamentals of Engineering Thermodynamics*, 9th ed., Moran, Shapiro, Boettner and Bailey (Zybooks version will be used)**

NOTE: MUST USE 9th Edition of textbook

Course Description:

1st Law of thermodynamics; work, heat and conversion of energy in closed and open systems, including engineering components, power and refrigeration cycles. Engineering problem-solving methodology, pure substance properties, property tables, equations of state, theory of gases and other states of matter. Second Law of thermodynamics; entropy, open and closed system entropy balances and equilibrium, with applications. Fundamental thermodynamic functions, Maxwell relations, Clapeyron equation.

Course Aims, Scope, Level:

The engineering science courses that form the basis of chemical engineering can be roughly divided into two sequences: transport processes and thermodynamics. This course is the first in the thermodynamics sequence and is designed to give the student a firm grounding in energy and entropy concepts. Later courses will deal in more detail with phase and reaction equilibrium. Knowledge of thermodynamics is needed in subsequent undergraduate chemical engineering courses. If a student can master the fundamentals early in the course of study, this knowledge will be of benefit throughout the rest of the curriculum as well as in a professional engineering career.

It is assumed that the students in the course have the necessary background in elementary mathematics, and this mathematical background will be used. Also, students should understand mechanics which is usually taught in the first college physics course.

By the end of the term, the students will understand energy and entropy concepts for closed systems and open systems (control volumes). Students will be able to apply engineering methodology to a variety of situations in order to solve thermodynamic problems. Energy and entropy are conceptually difficult and require time and effort on the student's part before they are well understood. Students should read the appropriate sections of the text before lectures, homework assignments should be started as soon as possible, and thoughtful questions should be brought up during lecture and discussion. Good luck!

Learning Objectives :

At the end of the course, students should be able to:

- Apply energy balances for closed systems and control volumes.
- Evaluate thermodynamic efficiencies of compressors, turbines, power plants, refrigeration and heat pump cycles.
- Identify phases at a given state and find thermodynamic properties such as internal energy, specific volumes from steam tables.
- Draw and interpret pressure-enthalpy (P-H), pressure-specific volume, temperature-specific volume, temperature-entropy charts.
- Relate thermodynamic properties via partial derivative, Maxwell's relations.
- Interpret phase diagrams of binary systems.
- Apply the entropy balance and identify if the process is reversible or irreversible.

Course Outline:

- I. Introduction
- II. Fundamentals (Chapter 1)
- III. The First Law (Chapter 2)
- IV. Properties of Pure Fluids (Chapter 3)
- V. Control Volume Analysis (Chapter 7)
- VI. The Second Law of Thermodynamics (Chapter 6)
- VII. Entropy (Chapter 4)
- VIII. Thermodynamic Relations (Chapter 11 Section 1 through 5)
- IX. Power Cycles (Chapter 9; select sections)

Exams, Homework and Grading

There will be 3 midterm exams, and one final exam. All exams are required. There will be two 1 hour 15 minute-lectures per week (M/W) and one, TA-led discussion session on F. New material will be covered in the lecture. Students should stay current with the material and be prepared to answer questions. During the discussion sessions, homework problems will be worked, and occasionally new material covered. The teaching assistant is under no obligation to grade homework that is late, illegible, or sloppy. Scoring fewer than half of the total homework grade is grounds for failing the course. The homework is an important exercise necessary for learning the material and should be worked with care. Students must submit HW via Gradescope. Exams will be held as paper copies during lecture time. After completion of the exams, instructor

will scan and upload them to Gradescope. Students will be able to review their graded HW and exams via Gradescope.

Quizzes

There will be weekly asynchronous quizzes. Each week, quiz will open on Blackboard at 12 pm on Friday and will remain open until 12 pm on Sunday.

Project

There will be one semester long project related to thermodynamics concept. Students will be working with groups and are expected to design a product with the focus of sustainability. Project includes research phase, modeling phase, prototype phase and testing phase. At the end of the semester, design competition will be held among all the groups. All the groups are required to create Adobe Express web pages for their design. Students will be graded based on understanding of the concept, design decisions, analysis of testing data and design constraints. Peer evaluations will be very important. Your peers in the project will be evaluating your performance. If your peer evaluation average is less than %75, your individual grade will be calculated by multiplying group grade by peer evaluation average percentage.

Attendance

Students are expected to attend all classes. Students with up to three absences will receive full (100%) attendance credit. Four absences will result in 75% attendance credit, five absences will result in 50% attendance credit, and six absences will result in 25% attendance credit. Students with more than six absences will receive a zero for attendance.

Class Activity

There will be 4 to 5 class activities through the semester. Students will be working with randomly selected groups on the problems given for that day. If any of the groups answers all the questions right, then everyone attended that day would receive extra credit.

Discussion Board

Microsoft Teams will be used as communication tool between students and instructor.

SaChE safety modules:

Students are expected to complete two SaChE safety modules online and submit their certificates of completion to the instructor. Students can choose two of the modules listed under Level One Curriculum on this website: <https://www.aiche.org/ccps/education/safety-and-chemical-engineering-education-sache-certificate-program>.

Grading Policy:

Course grades will be weighted according to the following grading formula.

Three unit-exams (10% each)	30%
Final exam (cumulative)	10%
Quizzes (weekly async.)	10%

Homework problems	15%
Project	15%
Attendance/Participation	15%
Sache Modules	5%
Kahoot (top-three place)(extra credit)	1%
Surveys (extra credit)	2%

The expectation is that all students will do well in this course if they actively participate in the course, do assignments on time and keep up with course materials.

Additional Information and Policies

1. EXAMS will be held during the class period. Makeup exams will not be provided except for health and weather-related emergencies.
2. Cheating on homework, quizzes, and exams will not be tolerated and appropriate action will be taken including (but not limited to) a failed grade or lowering of the final grade earned in the course.
3. Quizzes will be held weekly via BB. There will be no makeup quizzes.
4. Late Homework Policy: Homework can be turned in 1 day late without excuse for up to 50% credit.
5. Homework/quiz grading: Lowest homework and quiz grade will be dropped.
6. All the Homework will be submitted via GradeScope. Exams and quizzes will be uploaded to GradeScope and Blackboard by instructor and students will be able to access their graded paper via GradeScope.

Academic Integrity Policy

As an academic community, UIC is committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. All members of the campus community—students, staff, faculty, and administrators—share the responsibility of insuring that these standards are upheld so that such an environment exists. Instances of academic misconduct by students will be handled pursuant to the Student Disciplinary Policy: <http://dos.uic.edu/community-standards/>

Academic Misconduct

Please see collaboration policies written above for each type of work for this class. In all cases, you cannot submit someone else's work (partial or complete) as your own. You are guilty of academic misconduct if you extend or receive any kind of unauthorized assistance. Absolutely no transfer of program code between students is permitted (paper or electronic) under any circumstance, and you may not solicit code from other people, family, friends, or internet resources. Other examples of academic misconduct include emailing your work to another student, copying-pasting from the internet, working in a group on an assignment that states it is individual work, and allowing a tutor, TA, or another individual to write an answer or code for you. Academic misconduct is unacceptable. Minimum penalty will be a zero score on that assignment or assessment, with a letter drop to your final grade in the class. I reserve the right to exercise a change in the way grades are calculated if I deem it is necessary. More than one academic misconduct offense in this course will result in failure of the class. This means if you took this course a previous semester and were found guilty of academic misconduct, a second

offense this semester will result in failure from the course. Similarly, if you have two offenses in the same semester, you will receive an F in course. Note that the Dean of Students will change your grade from W to an F and only with my approval can it remain as a W (if you were found guilty and drop the class). EVERY academic misconduct offense must be reported to the Dean of Students (if you ask me not to, you are asking me to not do my job, so please don't).

Religious Holidays

Students who wish to observe their religious holidays shall notify the faculty member by the tenth day of the semester of the date when they will be absent unless the religious holiday is observed on or before the tenth day of the semester. In such cases, the student shall notify the faculty member at least five days in advance of the date when he/she will be absent. The faculty member shall make every reasonable effort to honor the request, not penalize the student for missing the class, and if an examination or project is due during the absence, give the student an exam or assignment equivalent to the one completed by those students in attendance. If the student feels aggrieved, he/she may request remedy through the campus grievance procedure: <http://policies.uic.edu/educational-policy/student-academic-grievance-policy/>

Disability Accommodation

The University of Illinois at Chicago is committed to maintaining a barrier-free environment so that students with disabilities can fully access programs, courses, services, and activities at UIC. Students with disabilities who require accommodations for access to and/or participation in this course are welcome, but must be registered with the Disability Resource Center (DRC). You may contact DRC at 312-413-2183 (v) or 773-649-4535 (VP/Relay) and consult the following: <http://drc.uic.edu/guide-to-accommodations>. If you have a letter of accommodation, it is best to stop by my office hours to talk about it in person. Any exams scheduled in the DRC must overlap with the originally scheduled exam. In other words, you must start the exam before or at the same time as the originally scheduled exam. Additionally, you may not leave the DRC exam room until after the originally scheduled exam is over or later.

Mental Health and Emotional Wellness

We value your mental health and emotional wellness as part of the UIC student experience. The UIC Counseling Center offers an array of services to provide additional support throughout your time at UIC, including workshops, peer support groups, counseling, self-help tools, and initial consultations to speak to a mental health counselor about your concerns. Please visit the Counseling Center website for more information (<https://counseling.uic.edu/>). Further, if you think emotional concerns may be impacting your academic success, please contact your faculty and academic advisers to create a plan to stay on track.

Campus Advocacy Network

If you have experienced sexual assault, domestic/dating violence, family abuse, stalking, or harassment, there is help available on campus. The Campus Advocacy Network provides free and confidential services to UIC students, faculty, and staff who have experienced or are experiencing interpersonal violence. If you would like to connect with an advocate, please contact CAN at can-appointment@uic.edu or call (312) 413-8206. To learn more, visit CAN's website at <https://can.uic.edu/>.